

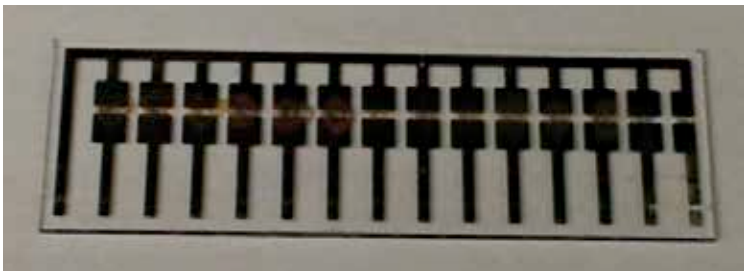
The intelligent use of sensors makes all the difference - it is what transforms a robot from a brute force capable merely of feats of strength or dexterity, into a responsive, interactive agent that can optimally handle a wide range of scenarios. This, coupled with the magic of data science and machine learning, is spurring robotics ahead like never before. In fact, this is precisely what allows the warehouse management robots used by giants like Amazon and Flipkart to accurately manage inventory completely autonomously. Autonomous behaviour - something that is set to save hundreds of thousands of man hours of labour - is what will eventually revolutionise the future of industry, and slowly phase out the working class. It remains to be seen if this will be a move for better or for worse.

How it obtains information - more on sensors

21st century research is ushering a grand new era of sensor technology. Things we never imagined could be converted into noiseless useful data are actually being measured and quantitatively studied for their applications to innovative products. The robotics sector has undoubtedly boosted the interest and funding pumped into research in this direction.

The standard implementations of most sensors are common knowledge - light dependent sensors (based on photodiodes or photovoltaic cells), acoustic sensors (which are just microphones), temperature sensors (implemented efficiently in integrated circuitry) and tactile sensors (starting from push buttons, but having graduated to touch foam and touch screens). The wave of unconventional sensors, on the other hand, begins to look like the technological foundations of a science fiction plot.

- **Sensors that measure distance:** ultrasonic/infrared proximity sensors and laser range-finders (which shoot out beams of light and measure



Source: <http://news.mt.edu>

This device, based on modified carbon nanotubes, can detect amines produced by decaying meat.